KAIZEN MATHS

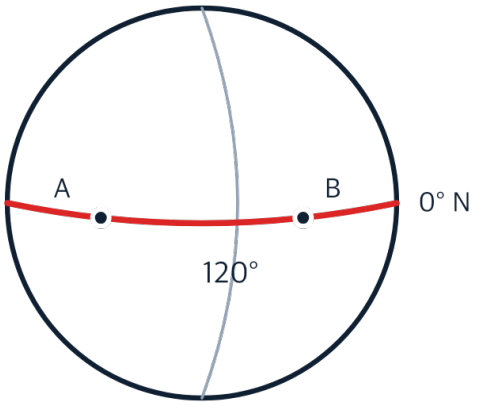
Lesson starter

Earth Geometry

Inverse and Time Problems

Name _____

SA1



Schematic globe; points lie on the shown latitude or meridian

Two places have longitudes that differ by 120°. Earth turns through 360° in 24 hours. Find the time difference between the two places.

GEO-F170-L3 · SA

Show your working

KAIZEN MATHS

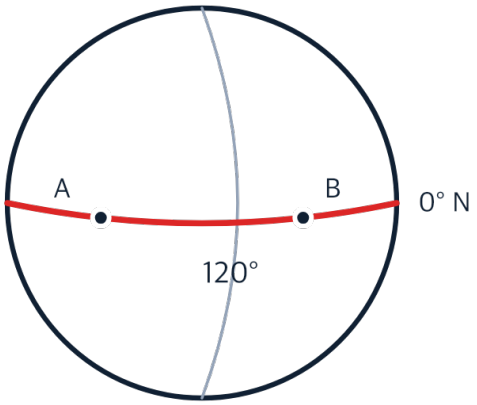
Lesson starter

Earth Geometry

Inverse and Time Problems

Name _____

SB1



Schematic globe; points lie on the shown latitude or meridian

The local time difference between two places is 8 hours. Earth turns 15° of longitude each hour. Find the longitude difference between the two places.

GEO-F170-L3 · SB

Show your working

KAIZEN MATHS

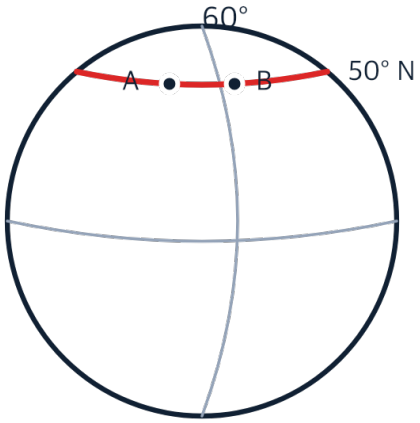
Lesson starter

Earth Geometry

Inverse and Time Problems

Name _____

SC1



Schematic globe; points lie on the shown latitude or meridian

A and B lie on latitude 50° N. The distance from A to B along the parallel is approximately 4,308 km. Taking R = 6400 km, find the longitude difference between A and B.

GEO-F170-L3 · SC

Show your working

KAIZEN MATHS

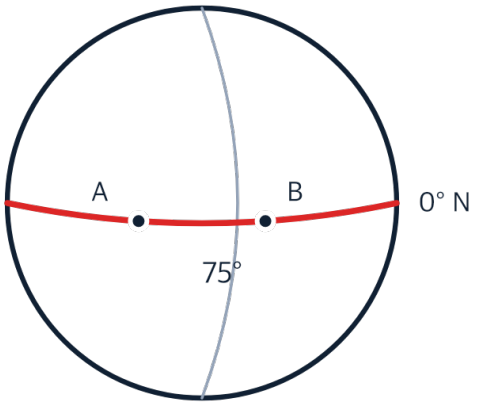
Lesson starter

Earth Geometry

Inverse and Time Problems

Name _____

SD1



Schematic globe; points lie on the shown latitude or meridian

Two places have longitudes that differ by 75°. Earth turns through 360° in 24 hours. Find the time difference between the two places.

GEO-F170-L3 · SD

Show your working

KAIZEN MATHS

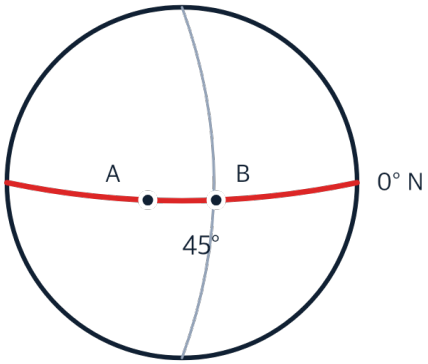
Exit ticket

Earth Geometry

Inverse and Time Problems

Name _____

EA1



Schematic globe; points lie on the shown latitude or meridian

The local time difference between two places is 3 hours. Earth turns 15° of longitude each hour. Find the longitude difference between the two places.

GEO-F170-L3 · EA

Show your working

KAIZEN MATHS

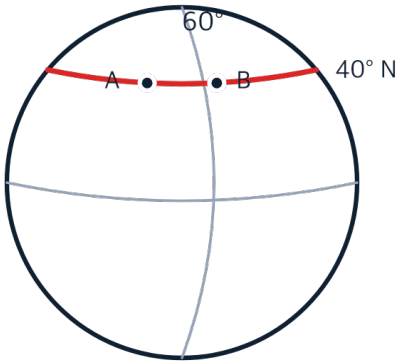
Exit ticket

Earth Geometry

Inverse and Time Problems

Name _____

EB1



Schematic globe; points lie on the shown latitude or meridian

A and B lie on latitude 40° N. The distance from A to B along the parallel is approximately 5,134 km. Taking R = 6400 km, find the longitude difference between A and B.

GEO-F170-L3 · EB

Show your working

KAIZEN MATHS

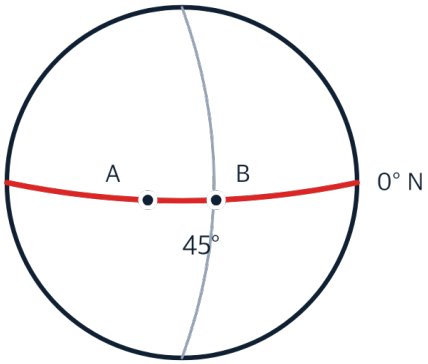
Exit ticket

Earth Geometry

Inverse and Time Problems

Name _____

EC1



Schematic globe; points lie on the shown latitude or meridian

Two places have longitudes that differ by 45°. Earth turns through 360° in 24 hours. Find the time difference between the two places.

GEO-F170-L3 · EC

Show your working

KAIZEN MATHS

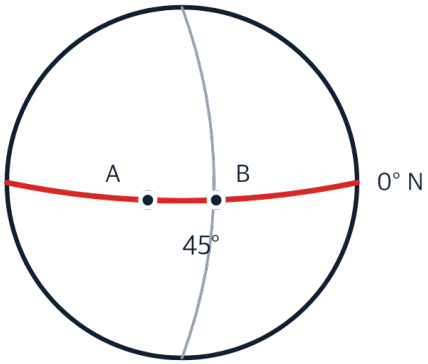
Exit ticket

Earth Geometry

Inverse and Time Problems

Name _____

EA1



Schematic globe; points lie on the shown latitude or meridian

The local time difference between two places is 3 hours. Earth turns 15° of longitude each hour. Find the longitude difference between the two places.

GEO-F170-L3 · EA

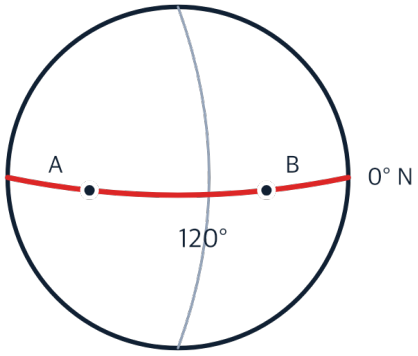
Show your working

Teacher answers & working

Earth Geometry · Inverse and Time Problems

GEO-F170-L3 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SA1



Schematic globe; points lie on the shown latitude or meridian

Two places have longitudes that differ by 120°. Earth turns through 360° in 24 hours. Find the time difference between the two places.

Answer:

8 hours

Working:

Step 1: Earth turns 360° in 24 hours.

Step 2: 360° ÷ 24 = 15° per hour.

Step 3: Divide the longitude difference by 15.

time difference = 120 ÷ 15

time difference = 8.00 hours

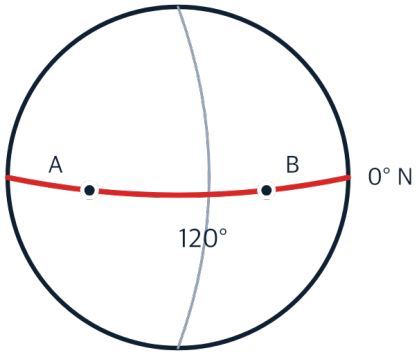
Final answer: 8 hours

Teacher answers & working

Earth Geometry · Inverse and Time Problems

GEO-F170-L3 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SB1



Schematic globe; points lie on the shown latitude or meridian

The local time difference between two places is 8 hours. Earth turns 15° of longitude each hour. Find the longitude difference between the two places.

Answer:

120°

Working:

Step 1: 1 hour corresponds to 15° of longitude.

Step 2: Multiply the time difference by 15.

longitude difference = 8.00 × 15

longitude difference = 120°

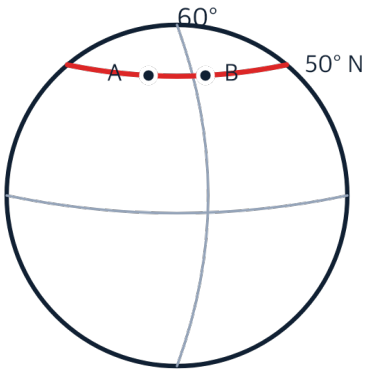
Final answer: 120°

Teacher answers & working

Earth Geometry · Inverse and Time Problems

GEO-F170-L3 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SC1



Schematic globe; points lie on the shown latitude or meridian

A and B lie on latitude 50° N. The distance from A to B along the parallel is approximately 4,308 km. Taking R = 6400 km, find the longitude difference between A and B.

Answer:

about 60°

Working:

Step 1: The route is along a parallel, so first find the small-circle radius.

small-circle radius = 6400 cos(50°) ≈ 4113.8 km

Step 2: Rearrange arc length: angle = (distance ÷ circumference) × 360.

angle ≈ (4308 ÷ (2π × 4113.8)) × 360

angle ≈ 60.0°

Step 3: Round to the nearest sensible longitude difference.

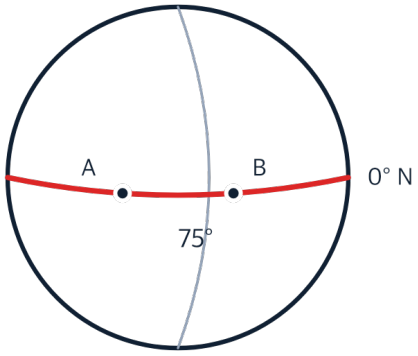
Final answer: about 60°

Teacher answers & working

Earth Geometry · Inverse and Time Problems

GEO-F170-L3 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SD1



Schematic globe; points lie on the shown latitude or meridian

Two places have longitudes that differ by 75°. Earth turns through 360° in 24 hours. Find the time difference between the two places.

Answer:

5 hours

Working:

Step 1: Earth turns 360° in 24 hours.

Step 2: 360° ÷ 24 = 15° per hour.

Step 3: Divide the longitude difference by 15.

time difference = 75 ÷ 15

time difference = 5.00 hours

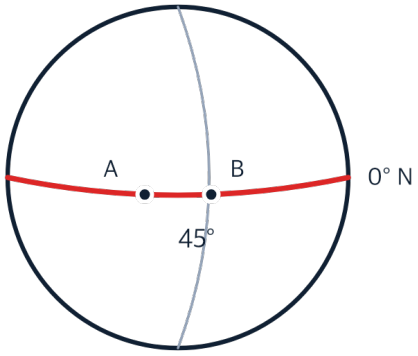
Final answer: 5 hours

Teacher answers & working

Earth Geometry · Inverse and Time Problems

GEO-F170-L3 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

EA1



Schematic globe; points lie on the shown latitude or meridian

The local time difference between two places is 3 hours. Earth turns 15° of longitude each hour. Find the longitude difference between the two places.

Answer:

45°

Working:

Step 1: 1 hour corresponds to 15° of longitude.

Step 2: Multiply the time difference by 15.

longitude difference = 3.00 × 15

longitude difference = 45°

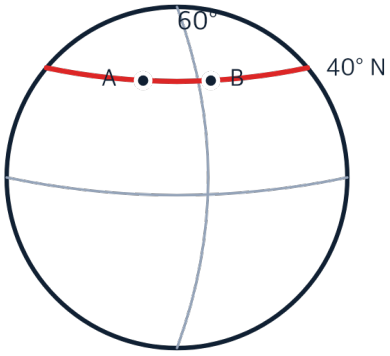
Final answer: 45°

Teacher answers & working

Earth Geometry · Inverse and Time Problems

GEO-F170-L3 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

EB1



Schematic globe; points lie on the shown latitude or meridian

A and B lie on latitude 40° N. The distance from A to B along the parallel is approximately 5,134 km. Taking R = 6400 km, find the longitude difference between A and B.

Answer:

about 60°

Working:

Step 1: The route is along a parallel, so first find the small-circle radius.

small-circle radius = 6400 cos(40°) ≈ 4902.7 km

Step 2: Rearrange arc length: angle = (distance ÷ circumference) × 360.

angle ≈ (5134 ÷ (2π × 4902.7)) × 360

angle ≈ 60.0°

Step 3: Round to the nearest sensible longitude difference.

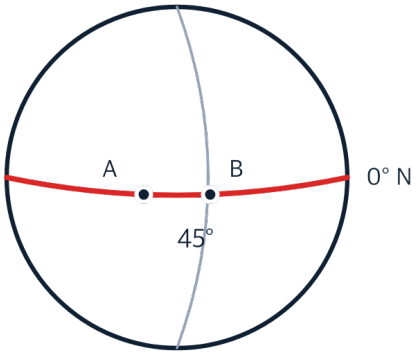
Final answer: about 60°

Teacher answers & working

Earth Geometry · Inverse and Time Problems

GEO-F170-L3 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

EC1



Schematic globe; points lie on the shown latitude or meridian

Two places have longitudes that differ by 45°. Earth turns through 360° in 24 hours. Find the time difference between the two places.

Answer:

3 hours

Working:

Step 1: Earth turns 360° in 24 hours.

Step 2: 360° ÷ 24 = 15° per hour.

Step 3: Divide the longitude difference by 15.

time difference = 45 ÷ 15

time difference = 3.00 hours

Final answer: 3 hours